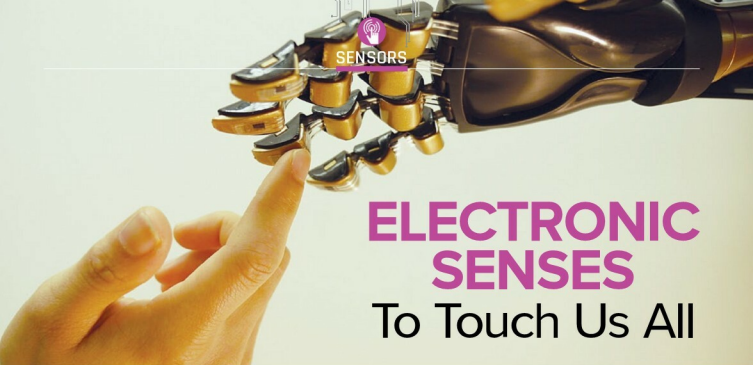




SENSORS

ELECTRONIC SENSES

To Touch Us All



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Electronics development has progressed well to replicate the human body's five basic senses with electronic devices called e-skin, e-tongue, e-nose, hearing aids and electronic eye. In fact, electronics has moved further to sixth and seventh senses with wearable interaction and gesture control of electronic devices.

Moving further into the digital world, we will be able to induce not just familiar but completely new, never-before-known senses. For example, a banking chip implanted under the skin may, one day, create the sense of a full or empty bank account. Connected to the social network, we will experience 'physically' the distance to those speaking, or the actuality and the time-freshness of what will be said.

The first bunch of digital senses will simulate natural ones, just because we have to rest on something familiar in our cognitive experience. Then, we will develop derivative senses; these derivations will detach farther and farther from what we have been familiar with in our previous physical experience.

Once in the future, we will have moved far enough to become completely resettled in the digital environment that will turn out to be the natural environment for the new, digital post-human being. By that moment, we will be able to design completely new sensations with whatever range of excitement.

Basics

In electronics, sense is a technique used in power supplies to produce the correct voltage for a load. Although simple batteries

naturally maintain a steady voltage (except in cases of large internal impedance), a power supply must use a feedback system to make adjustments based on the difference between its intended output and actual output. If this system is working, the actual output will be very close to the intended output.

Two types of senses are used depending on where the power supply output is measured. In local sense, the supply simply measures the voltage at its output terminals, where the leads connect them to the load. This method has the problem of not accounting for the voltage drop due to resistance of the leads, which is proportional to the amount of current drawn by the load. That is, the supply might be producing the correct voltage at its output terminals, but there will be a lower voltage at the input terminals of the load.

When this might cause a problem, remote sense can be used to force the power supply to counteract the voltage drop by raising the voltage at its output terminals. If successful, it will exactly cancel the drop along the leads, yielding the correct voltage at the input terminals of the load. This is accomplished by using separate sense leads, which are connected to the load's input terminals, to measure the output voltage. (Because the sensing function draws only a very small amount of current, there is practically no additional voltage drop due to the sense leads themselves.)

Many power supplies that are equipped with remote sense can cause catastrophic damage to the loads if these turn on while the sense leads are unconnected. To avoid this, some supplies are equipped with auto sense,

Electronic senses offer advantages of consistency as well as not having to use humans for what may sometimes be unpleasant or even dangerous tasks